

Additive Moment Rigidity and the Alaoglu–Erdős Problem

A Hausdorff–Müntz continuation beyond the supplied unified report

Independent research report

August 21, 2026

Status statement. The implication

$$2^x, 3^x \in \mathbb{Z} \implies x \in \mathbb{Z}$$

is not proved in this report. As of the date above it remains open, and a positive answer follows from the Four Exponentials Conjecture. This document nevertheless gives a detailed, nonduplicative investigation of a new route based on the additive relation $3 = 2 + 1$. All theorems labeled as such below are proved. The main outcome is both constructive and negative: a hypothetical counterexample produces an explicit Hausdorff moment sequence with several rational moments and exact prime-power denominators, but finite rational quadrature and a Müntz incompleteness theorem show rigorously why linear moment positivity cannot finish the problem. The remaining step is isolated as a nonlinear arithmetic-rigidity target.

Abstract

Let $x \in \mathbb{R}$ satisfy $2^x, 3^x \in \mathbb{Z}$, and suppose x is not an integer. Writing $x = n + \alpha$ with $0 < \alpha < 1$ gives rational numbers $u = 2^\alpha \in \mathbb{Z}[1/2]$ and $v = 3^\alpha \in \mathbb{Z}[1/3]$. The supplied unified report already pushes transcendence, determinant, p -adic, matrix, probability, symbolic-dynamical, and polynomial-approximation methods to their current barriers. The present report instead exploits $3 = 2 + 1$ by studying the increments

$$d_j = (j + 2)^\alpha - (j + 1)^\alpha.$$

They are the moments of the explicit positive measure

$$d\mu_\alpha(q) = \frac{\alpha}{\Gamma(1 - \alpha)} (1 - q)(-\log q)^{-\alpha-1} dq, \quad 0 < q < 1.$$

Consequently (d_j) is strictly completely monotone, all its Hankel matrices are positive definite, and it is strictly log-convex. The consecutive 3-smooth pairs are exactly $(1, 2), (2, 3), (3, 4), (8, 9)$, so a counterexample makes precisely d_0, d_1, d_2, d_7 immediately rational. We compute exact reduced denominators for the first Hankel defect and three higher log-convexity defects; the numerators are positive odd integers and are also prime to 3 whenever a triadic denominator is present.

The decisive audit is then twofold. First, every finite set of these rational moment constraints has a positive atomic realization with rational nodes and rational weights; an explicit two-atom formula already reproduces d_0, d_1, d_2 . Second, the exponents $2^a 3^b - 1$ have a convergent reciprocal sum. The full Müntz theorem therefore supplies bounded, compactly supported perturbations of μ_α that preserve *all* linear 3-smooth observations

$$\int_0^1 \frac{1 - q^{N-1}}{1 - q} d\mu(q), \quad N = 2^a 3^b,$$

while leaving the endpoint singularities and moment asymptotics unchanged. Thus no proof can come from linear positivity, finite Hankel certificates, measure reconstruction, or endpoint tail data alone. A successful argument must use nonlinear coupling between the explicit one-parameter density, multiplicativity, and the separated dyadic/triadic denominator arithmetic. That exact target is formulated at the end.

Contents

1	Scope, current status, and novelty	1
1.1	The problem	1
1.2	Relation to the supplied unified report	1
2	Minimal reductions and the four-exponentials wall	1
3	Fractional normalization and prime-separated denominators	2
4	The Hausdorff moment representation	3
4.1	The increment sequence	3
4.2	Complete monotonicity, Hankel positivity, and log-convexity	4
4.3	Tail behavior	5

5	Exactly which adjacent 3-smooth values are rational?	5
6	Exact prime-power denominators of moment defects	6
6.1	Arithmetic normalization	6
6.2	The first Hankel defect	6
6.3	Three higher log-convexity defects	7
6.4	What the exact defects do and do not buy	8
7	Finite rational quadrature: a no-go theorem	8
7.1	An explicit two-atom model	8
7.2	A general rational finite-observable theorem	9
8	The complete 3-smooth observation family	10
8.1	The observation kernels	10
8.2	Why a density theorem is the right audit	11
9	Müntz invisibility: an infinite-dimensional no-go theorem	11
10	An analytic reformulation of the invisible directions	12
11	What would constitute a proof along this route?	13
11.1	The exact nonlinear target	13
11.2	Why the current exact defects do not close	14
11.3	A precise conditional resolution	14
12	Comparison with the supplied report	14
13	Conclusion	15
A	Algebraic audit of the exact denominator formulas	16
B	A moment-cone interpretation	16
C	Classical inputs and bibliography notes	17

1 Scope, current status, and novelty

1.1 The problem

The Alaoglu–Erdős integer-exponent conjecture is the following elementary-looking statement.

Conjecture 1.1 (Alaoglu–Erdős). *If $x \in \mathbb{R}$ and $2^x, 3^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

The question appears in the circle of problems originating in Alaoglu and Erdős’s work on highly composite and colossally abundant numbers [1]. Waldschmidt’s modern survey states the question explicitly and records that a positive answer follows from the four exponentials problem; the latter remains open [11]. Recent work on matrices of logarithms likewise formulates, rather than resolves, a matrix-coefficient conjecture designed to control vanishing logarithmic determinants [5].

The statement is easy for rational or algebraic exponents. Its difficulty is that any counterexample must be transcendental and creates a singular 2×2 matrix of logarithms at exactly the size not reached by the Six Exponentials Theorem. This is not a reason to stop; it is a constraint on what a valid proof must add.

1.2 Relation to the supplied unified report

The supplied source `Article/alaoglu-erdos-unified-report/alaoglu-erdos-unified-report.tex` is a very large audit of the existing project. It already develops:

- the reduction to Four Exponentials and the Gelfond–Schneider consequences;
- exact conditional free-monoid structure if one counterexample exists;
- ordinary and generalized Vandermonde determinants, interpolation, Loewner matrices, HCIZ asymptotics, and finite differences;
- p -adic valuation cascades, tropical assignment layers, and formal Lean modules;
- rational-point counting, Sturmian and automatic-sequence models, probability, information-theoretic, Mahler, spectral, and polynomial-approximation diagnostics.

This report does not repeat those developments. It begins only with the minimal reductions needed for a standalone argument, and then studies a direction absent from the supplied text: *the Hausdorff moment structure forced by the additive adjacency of the 3-smooth numbers 1, 2, 3, 4, 8, 9*. The strongest conclusions are new exact denominator theorems and two no-go theorems—finite rational quadrature and infinite Müntz invisibility.

Trust boundary. The report proves every displayed theorem by ordinary mathematical arguments. It does not claim a Lean formalization. Classical inputs used as black boxes are Gelfond–Schneider, Catalan–Mihăilescu, and the full Müntz theorem in L^1 on an interval away from the origin.

2 Minimal reductions and the four-exponentials wall

Proposition 2.1 (Elementary and transcendence reductions). *Suppose $2^x, 3^x \in \mathbb{Z}$.*

- (i) $x \geq 0$.
- (ii) If $x \in \mathbb{Q}$, then $x \in \mathbb{N}_0$.
- (iii) If x is algebraic, then $x \in \mathbb{N}_0$.
- (iv) Assuming the Four Exponentials Conjecture, $x \in \mathbb{N}_0$.

Consequently every hypothetical counterexample is a positive transcendental irrational.

Proof. If $x < 0$, then $0 < 2^x < 1$, so 2^x cannot be an integer. This proves (i).

For (ii), write $x = a/b$ in lowest terms with $a \geq 0$ and $b \geq 1$. If $M = 2^x \in \mathbb{N}$, then $M^b = 2^a$. Taking the 2-adic valuation gives $b v_2(M) = a$. Hence $b \mid a$; coprimality forces $b = 1$.

For (iii), if x were algebraic and irrational, the Gelfond–Schneider theorem would make 2^x transcendental, contradicting $2^x \in \mathbb{Z}$. Part (ii) handles the remaining algebraic case.

For (iv), assume x is not rational. The pair $(\log 2, \log 3)$ is linearly independent over $\mathbb{Q}$, because $2^a 3^b = 1$ with $a, b \in \mathbb{Z}$ forces $a = b = 0$. The pair $(1, x)$ is also $\mathbb{Q}$ -linearly independent. The four numbers

$$e^{(\log 2) \cdot 1} = 2, \quad e^{(\log 2)x} = 2^x, \quad e^{(\log 3) \cdot 1} = 3, \quad e^{(\log 3)x} = 3^x$$

are algebraic. Four Exponentials would force one to be transcendental, a contradiction. Thus $x \in \mathbb{Q}$, and (ii) applies. $\square$

Equivalently, a counterexample gives positive integers M, A with

$$\frac{\log M}{\log 2} = \frac{\log A}{\log 3} = x, \quad \det \begin{pmatrix} \log 2 & \log 3 \\ \log M & \log A \end{pmatrix} = 0. \quad (1)$$

The row and column pairs are $\mathbb{Q}$ -independent when x is irrational. Equation (1) is therefore a concrete 2×2 instance of the four-exponentials barrier. The rest of this report asks whether the special additive relation $3 = 2 + 1$ adds something that the logarithmic determinant does not see.

3 Fractional normalization and prime-separated denominators

Let x be a hypothetical counterexample. Put

$$n = \lfloor x \rfloor, \quad \alpha = x - n \in (0, 1), \quad u = 2^\alpha, \quad v = 3^\alpha. \quad (2)$$

Since 2^x and 3^x are integers,

$$u = \frac{2^x}{2^n} \in \mathbb{Q}, \quad v = \frac{3^x}{3^n} \in \mathbb{Q}.$$

Write these fractions in lowest terms as

$$u = \frac{P}{2^r}, \quad v = \frac{Q}{3^s}, \quad (3)$$

where

$$1 \leq r \leq n, \quad 0 \leq s \leq n, \quad P \text{ is odd}, \quad 3 \nmid Q. \quad (4)$$

The inequality $r \geq 1$ follows from $1 < u < 2$: a rational integer strictly between 1 and 2 does not exist. The case $s = 0$ is not excluded a priori, because the integer $v = 2$ lies in $(1, 3)$.

Define the 3-smooth semigroup

$$\mathcal{S}_{2,3} = \{2^a 3^b : a, b \in \mathbb{N}_0\}.$$

For every $N = 2^a 3^b \in \mathcal{S}_{2,3}$,

$$N^\alpha = u^a v^b \in \mathbb{Q}. \quad (5)$$

Thus one hypothetical solution gives infinitely many rational values of the concave function $t \mapsto t^\alpha$ —but only on a logarithmically sparse multiplicative semigroup.

Two elementary endpoint gaps will be useful for calibration.

Lemma 3.1 (The one-coordinate endpoint gaps). *With the notation above,*

$$\alpha \geq 2^{-r}, \quad 1 - \alpha \geq \frac{2^{-r}}{2 \log 2}. \quad (6)$$

In particular both distances are bounded below exponentially in n .

Proof. The reduced numerator of $u - 1 = (P - 2^r)/2^r$ is a positive odd integer, so $u - 1 \geq 2^{-r}$. Convexity of 2^t on $[0, 1]$ places its graph below the chord joining $(0, 1)$ and $(1, 2)$; hence $2^\alpha - 1 \leq \alpha$. This proves the first inequality.

Similarly, $2 - u = (2^{r+1} - P)/2^r \geq 2^{-r}$. Also

$$2 - 2^\alpha = \int_\alpha^1 2^t \log 2 \, dt \leq 2 \log 2 (1 - \alpha).$$

Combining the two bounds proves the second inequality. $\square$

These simple gaps are important when evaluating later determinant bounds: a new positivity defect is only useful if it beats information already visible in the single dyadic coordinate.

4 The Hausdorff moment representation

4.1 The increment sequence

For $j \in \mathbb{N}_0$ define

$$d_j = (j + 2)^\alpha - (j + 1)^\alpha. \quad (7)$$

Because $0 < \alpha < 1$, the sequence is positive and strictly decreasing. More is true: it is a strict Hausdorff moment sequence with an explicit density.

Theorem 4.1 (Explicit Hausdorff measure). *For $0 < \alpha < 1$, define a measure on $(0, 1)$ by*

$$d\mu_\alpha(q) = w_\alpha(q) \, dq, \quad w_\alpha(q) = \frac{\alpha}{\Gamma(1 - \alpha)} (1 - q)(-\log q)^{-\alpha-1}. \quad (8)$$

Then μ_α is finite, positive, and has full support in $[0, 1]$. Moreover

$$d_j = \int_0^1 q^j \, d\mu_\alpha(q) \quad (j \in \mathbb{N}_0). \quad (9)$$

More generally, for every real $z > -1$,

$$\int_0^1 q^z d\mu_\alpha(q) = (z+2)^\alpha - (z+1)^\alpha. \quad (10)$$

Proof. For $t > 0$, the gamma integral gives

$$t^{\alpha-1} = \frac{1}{\Gamma(1-\alpha)} \int_0^\infty e^{-ts} s^{-\alpha} ds.$$

Since

$$d_j = \int_{j+1}^{j+2} \alpha t^{\alpha-1} dt,$$

Tonelli's theorem yields

$$\begin{aligned} d_j &= \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty s^{-\alpha} \left(\int_{j+1}^{j+2} e^{-ts} dt \right) ds \\ &= \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty e^{-js} e^{-s} (1 - e^{-s}) s^{-\alpha-1} ds. \end{aligned}$$

Put $q = e^{-s}$. Then $ds = -dq/q$, and the last expression becomes

$$\frac{\alpha}{\Gamma(1-\alpha)} \int_0^1 q^j (1-q) (-\log q)^{-\alpha-1} dq,$$

which is (9).

Near $q = 1$, $-\log q \sim 1 - q$, so the density is asymptotic to a positive constant times $(1 - q)^{-\alpha}$ and is integrable because $\alpha < 1$. Near $q = 0$ the change of variables $q = e^{-s}$ shows integrability. The density is strictly positive on $(0, 1)$, hence the support is all of $[0, 1]$. Replacing the integer j by any real $z > -1$ in the same computation gives (10). $\square$

4.2 Complete monotonicity, Hankel positivity, and log-convexity

Corollary 4.2 (Strict complete monotonicity). *For all $j, k \in \mathbb{N}_0$,*

$$(-1)^k \Delta^k d_j = \int_0^1 q^j (1-q)^k d\mu_\alpha(q) > 0, \quad (11)$$

where $\Delta d_j = d_{j+1} - d_j$.

Proof. Apply Δ^k to q^j inside (9). Since $\Delta^k q^j = q^j (q-1)^k$, the formula follows. Strict positivity uses the positive density on $(0, 1)$. $\square$

Corollary 4.3 (Strict Hankel positivity). *For every $m \geq 0$, the Hankel matrix*

$$H_m = (d_{i+j})_{0 \leq i, j \leq m}$$

is positive definite. In particular,

$$d_0 d_2 - d_1^2 > 0. \quad (12)$$

Proof. For a nonzero real vector $(c_0, \dots, c_m)$,

$$\sum_{i,j=0}^m c_i c_j d_{i+j} = \int_0^1 \left(\sum_{i=0}^m c_i q^i \right)^2 d\mu_\alpha(q) > 0.$$

A nonzero polynomial cannot vanish on the full support of μ_α . □

Corollary 4.4 (Strict log-convexity). *If $0 \leq i < j < k$, then*

$$d_j^{k-i} < d_i^{k-j} d_k^{j-i}. \quad (13)$$

Proof. Write

$$j = \frac{k-j}{k-i} i + \frac{j-i}{k-i} k$$

and apply Hölder's inequality to $q^j = (q^i)^{(k-j)/(k-i)} (q^k)^{(j-i)/(k-i)}$. Equality would force q^i and q^k to be proportional almost everywhere, impossible on a measure with full support. □

4.3 Tail behavior

The mean-value theorem gives

$$d_j = \alpha \xi_j^{\alpha-1} \quad \text{for some } \xi_j \in (j+1, j+2), \quad d_j \sim \alpha j^{\alpha-1}. \quad (14)$$

Thus the moment decay is polynomial, not exponential. In measure language, this is caused by the integrable singularity of w_α at $q = 1$. This observation will separate the genuine infinite moment sequence from the finite atomic models constructed later.

5 Exactly which adjacent 3-smooth values are rational?

The formula $d_j = (j+2)^\alpha - (j+1)^\alpha$ is rational whenever both consecutive integers $j+1$ and $j+2$ are 3-smooth. Those pairs can be classified exactly.

Theorem 5.1 (Consecutive 3-smooth pairs). *The positive consecutive pairs $(N, N+1)$ for which both entries belong to $\mathcal{S}_{2,3}$ are*

$$(1, 2), \quad (2, 3), \quad (3, 4), \quad (8, 9). \quad (15)$$

Proof. The pair $(1, 2)$ is immediate. Suppose $N > 1$. Since $\gcd(N, N+1) = 1$ and both numbers have all prime factors in $\{2, 3\}$, their prime supports are disjoint. Hence one is a pure power of 2 and the other a pure power of 3:

$$\{N, N+1\} = \{2^a, 3^b\} \quad (a, b \geq 1).$$

If $a, b > 1$, Catalan–Mihăilescu says that the unique pair of nontrivial perfect powers that differ by 1 is 8 and 9 [8]. If one exponent equals 1, direct inspection gives 2, 3 and 3, 4. No other cases occur. □

Corollary 5.2 (The four immediately rational moments). *Under the hypothetical counterexample normalization (2),*

$$\begin{aligned} d_0 &= u - 1, \\ d_1 &= v - u, \\ d_2 &= u^2 - v, \\ d_7 &= v^2 - u^3, \end{aligned} \tag{16}$$

so $d_0, d_1, d_2, d_7 \in \mathbb{Q}_{>0}$.

The list is exact for unit increments. Of course every 3-smooth value gives a rational *partial sum* of the moments:

$$N^\alpha - 1 = \sum_{j=0}^{N-2} d_j = \int_0^1 \frac{1 - q^{N-1}}{1 - q} d\mu_\alpha(q), \quad N \in \mathcal{S}_{2,3}, \tag{17}$$

but only the four pairs in (15) isolate single moments directly.

6 Exact prime-power denominators of moment defects

6.1 Arithmetic normalization

Retain (3). Define the positive integers

$$\begin{aligned} A_0 &= P - 2^r, \\ A_1 &= 2^r Q - 3^s P, \\ A_2 &= 3^s P^2 - 2^{2r} Q, \\ A_7 &= 2^{3r} Q^2 - 3^{2s} P^3. \end{aligned} \tag{18}$$

Then

$$d_0 = \frac{A_0}{2^r}, \quad d_1 = \frac{A_1}{2^r 3^s}, \quad d_2 = \frac{A_2}{2^{2r} 3^s}, \quad d_7 = \frac{A_7}{2^{3r} 3^{2s}}. \tag{19}$$

Every A_i is odd. If $s \geq 1$, then

$$3 \nmid A_1 A_2 A_7. \tag{20}$$

Indeed, the four numerators in (18) are respectively odd-minus-even, even-minus-odd, odd-minus-even, and even-minus-odd. Modulo 3, when $s \geq 1$,

$$A_1 \equiv 2^r Q, \quad A_2 \equiv -2^{2r} Q, \quad A_7 \equiv 2^{3r} Q^2,$$

all nonzero.

6.2 The first Hankel defect

Theorem 6.1 (Exact denominator of the first Hankel defect). *Let*

$$\mathcal{H} = d_0 d_2 - d_1^2. \tag{21}$$

Then

$$\mathcal{H} = (u - 1)(u^2 - v) - (v - u)^2 = u^3 - 2u^2 + uv + v - v^2 > 0, \tag{22}$$

and in lowest terms

$$\mathcal{H} = \frac{3^s A_0 A_2 - 2^r A_1^2}{2^{3r} 3^{2s}}. \quad (23)$$

In particular, the reduced denominator is exactly $2^{3r} 3^{2s}$ and

$$\mathcal{H} \geq 2^{-3r} 3^{-2s}. \quad (24)$$

Proof. Positivity is Corollary 4.3; the expression in u, v is a direct expansion. Substitution of (19) gives (23). Its numerator is odd because the first term is odd and the second is even. If $s \geq 1$, the numerator is congruent modulo 3 to $-2^r A_1^2 \not\equiv 0$. If $s = 0$, no factor of 3 occurs in the denominator. Hence no factor 2 or 3 cancels, proving exactness. The positive numerator is at least 1. $\square$

The double-integral identity

$$\mathcal{H} = \frac{1}{2} \int_0^1 \int_0^1 (q - t)^2 d\mu_\alpha(q) d\mu_\alpha(t) \quad (25)$$

shows that $\mathcal{H}$ is the unnormalized variance of the Hausdorff measure. Theorem 6.1 therefore turns a continuous variance into a positive integer divided by a fully determined prime-power denominator.

6.3 Three higher log-convexity defects

Apply Corollary 4.4 to the triples $(0, 1, 7)$, $(1, 2, 7)$, and $(0, 2, 7)$.

Theorem 6.2 (Exact higher defects). *The rational numbers*

$$\mathcal{F}_{017} = d_0^6 d_7 - d_1^7, \quad (26)$$

$$\mathcal{F}_{127} = d_1^5 d_7 - d_2^6, \quad (27)$$

$$\mathcal{F}_{027} = d_0^5 d_7^2 - d_2^7 \quad (28)$$

are strictly positive. Their exact reduced forms are

$$\mathcal{F}_{017} = \frac{3^{5s} A_0^6 A_7 - 2^{2r} A_1^7}{2^{9r} 3^{7s}}, \quad (29)$$

$$\mathcal{F}_{127} = \frac{2^{4r} A_1^5 A_7 - 3^s A_2^6}{2^{12r} 3^{7s}}, \quad (30)$$

$$\mathcal{F}_{027} = \frac{2^{3r} 3^{3s} A_0^5 A_7^2 - A_2^7}{2^{14r} 3^{7s}}. \quad (31)$$

Each displayed numerator is a positive odd integer; if $s \geq 1$, it is coprime to 6. In every case it is coprime to the displayed denominator. Consequently

$$\mathcal{F}_{017} \geq 2^{-9r} 3^{-7s}, \quad \mathcal{F}_{127} \geq 2^{-12r} 3^{-7s}, \quad \mathcal{F}_{027} \geq 2^{-14r} 3^{-7s}. \quad (32)$$

Proof. Strict positivity is exactly (13) for the three indicated index triples. Equations (29)–(31) follow by substituting (19) and taking a common denominator.

For (29), the first numerator term is odd and the second even, so the numerator is odd. If $s \geq 1$, it is congruent to $-2^{2r} A_1^7 \not\equiv 0 \pmod{3}$. For (30), the first term is even and the second odd, and modulo 3 the numerator is $2^{4r} A_1^5 A_7 \not\equiv 0$. For (31), the first term is even and the second odd, and modulo 3 the numerator is $-A_2^7 \not\equiv 0$. The case $s = 0$ again has no triadic denominator. Exactness and the lower bounds follow. $\square$

Table 1: Exact arithmetic defects supplied by the four rational moments.

Defect	Positivity source	Exact denominator	Numerator information
$d_0 d_2 - d_1^2$	2×2 Hankel	$2^{3r} 3^{2s}$	positive; coprime to the denominator
$d_0^6 d_7 - d_1^7$	log-convex $(0, 1, 7)$	$2^{9r} 3^{7s}$	positive; coprime to the denominator
$d_1^5 d_7 - d_2^6$	log-convex $(1, 2, 7)$	$2^{12r} 3^{7s}$	positive; coprime to the denominator
$d_0^5 d_7^2 - d_2^7$	log-convex $(0, 2, 7)$	$2^{14r} 3^{7s}$	positive; coprime to the denominator

6.4 What the exact defects do and do not buy

The exact denominator is genuine arithmetic information, but its immediate size consequence is not enough. For example, since $d_7 < d_0$ and $d_0 = 2^\alpha - 1 \leq \alpha$,

$$0 < \mathcal{F}_{017} < d_0^7 \leq \alpha^7.$$

Combining this with (32) gives

$$\alpha \geq 2^{-9r/7} 3^{-s}. \quad (33)$$

This is weaker than the elementary dyadic estimate $\alpha \geq 2^{-r}$ in Lemma 3.1 for many parameter ranges, and in any case remains exponentially small in n . Similarly, an upper bound for $\mathcal{H}$ near $\alpha = 1$ yields only another exponential endpoint gap. The defects therefore provide exact certificates and useful formalization targets, but not a contradiction.

This failure motivates the next question: can one combine many moment inequalities? The answer is unexpectedly rigid in the opposite direction. Every finite collection can be simulated by a rational atomic measure.

7 Finite rational quadrature: a no-go theorem

7.1 An explicit two-atom model

Theorem 7.1 (Rational two-atom reproduction). *Let $m_0, m_1, m_2 \in \mathbb{Q}_{>0}$ satisfy*

$$m_0 > m_1 > m_2, \quad m_0 m_2 > m_1^2.$$

Then the positive atomic measure

$$\nu = \left(m_0 - \frac{m_1^2}{m_2} \right) \delta_0 + \frac{m_1^2}{m_2} \delta_{m_2/m_1} \quad (34)$$

has rational nodes and rational weights and satisfies

$$\int q^j d\nu(q) = m_j \quad (j = 0, 1, 2). \quad (35)$$

In particular, under a hypothetical Alaoglu–Erdős counterexample, the first three moments d_0, d_1, d_2 have a completely rational positive model unrelated to the transcendental parameter α .

Proof. The node $t = m_2/m_1$ lies in $(0, 1)$. The weight at 0 is $(m_0 m_2 - m_1^2)/m_2 > 0$, and the other weight is positive. Direct calculation gives

$$\nu([0, 1]) = m_0, \quad \int q \, d\nu = \frac{m_1^2 m_2}{m_2 m_1} = m_1, \quad \int q^2 \, d\nu = \frac{m_1^2 m_2^2}{m_2 m_1^2} = m_2.$$

All quantities are rational. □

For $m_j = d_j$, the positivity of the first weight is exactly the Hankel inequality $d_0 d_2 - d_1^2 > 0$. Thus the first nontrivial Hankel certificate is not merely compatible with rational data; it canonically constructs a rational quadrature formula for it.

7.2 A general rational finite-observable theorem

The preceding phenomenon is not confined to three moments.

Lemma 7.2 (Rational interior quadrature). *Let $f_1, \dots, f_m \in \mathbb{Q}[q]$ be linearly independent real polynomials, and let μ be a finite positive measure with a strictly positive density on $(0, 1)$. Suppose*

$$y_i = \int_0^1 f_i(q) \, d\mu(q) \in \mathbb{Q} \quad (1 \leq i \leq m).$$

Then there are finitely many rational nodes $q_1, \dots, q_L \in (0, 1)$ and positive rational weights $w_1, \dots, w_L$ such that

$$y_i = \sum_{\ell=1}^L w_\ell f_i(q_\ell) \quad (1 \leq i \leq m). \quad (36)$$

Proof. Set $F(q) = (f_1(q), \dots, f_m(q)) \in \mathbb{R}^m$, and let C be the convex cone generated by $F((0, 1))$. The vector $y = (y_1, \dots, y_m)$ lies in the interior of C . Indeed, if a nonzero linear functional $c \in (\mathbb{R}^m)^*$ supported C at y , then the polynomial $p(q) = c \cdot F(q)$ would be nonnegative on $(0, 1)$ and satisfy $\int p \, d\mu = 0$. Positivity of the density would force p to vanish identically, contrary to linear independence of the f_i .

By finite-dimensional convex geometry, an interior point of a generated cone lies in the interior of a cone generated by finitely many of its generating rays. Choose such nodes and then perturb them slightly to distinct rational nodes. Interior containment is open, so it persists under a sufficiently small perturbation. We obtain a rational matrix $B = (f_i(q_\ell))$ for which the system $Bw = y$ has a strictly positive real solution. The affine solution space is defined over $\mathbb{Q}$; its rational points are dense. Hence it has a strictly positive rational solution, which gives (36). □

Corollary 7.3 (Finite simulation of all available additive data). *Assume a hypothetical counterexample. Then:*

- (i) *the four rational moments d_0, d_1, d_2, d_7 are reproduced exactly by a finite positive measure with rational atoms and rational weights;*
- (ii) *for every finite set $N_1, \dots, N_m \in \mathcal{S}_{2,3}$, there is such a rational atomic measure ν satisfying*

$$\int_0^1 \frac{1 - q^{N_i-1}}{1 - q} \, d\nu(q) = N_i^\alpha - 1 \quad (1 \leq i \leq m).$$

Proof. For (i), use Lemma 7.2 with $f(q) = 1, q, q^2, q^7$ and (16). For (ii), use the linearly independent polynomials

$$K_{N_i-1}(q) = \frac{1 - q^{N_i-1}}{1 - q} = 1 + q + \cdots + q^{N_i-2}$$

after deleting repetitions. Their integrals are rational by (17) and (5). $\square$

Warning 7.4 (Finite positivity cannot prove the conjecture). Any argument that uses only a finite number of linear moment identities and positivity of integrals against nonnegative polynomials can be reproduced inside a completely rational finite atomic model. Therefore no finite Hankel matrix, no finite list of complete-monotonicity inequalities, and no finite positive-kernel certificate can distinguish the putative counterexample from rational quadrature data. A proof must be genuinely infinite or use additional nonlinear arithmetic.

8 The complete 3-smooth observation family

8.1 The observation kernels

For $m \geq 1$, put

$$K_m(q) = \frac{1 - q^m}{1 - q} = 1 + q + \cdots + q^{m-1}. \quad (37)$$

Equation (17) becomes

$$\int_0^1 K_{N-1}(q) d\mu_\alpha(q) = N^\alpha - 1 \quad (N \in \mathcal{S}_{2,3}, N \geq 2). \quad (38)$$

Under a counterexample, the right side is not merely rational; it factors coherently as $u^a v^b - 1$ when $N = 2^a 3^b$.

Let

$$\Lambda = \{N - 1 : N \in \mathcal{S}_{2,3}, N \geq 2\}. \quad (39)$$

This exponent set is too thin for Müntz completeness.

Lemma 8.1 (Convergent reciprocal sum). *The set Λ satisfies*

$$\sum_{m \in \Lambda} \frac{1}{m} < \infty. \quad (40)$$

More explicitly, the sum is at most 4.

Proof. For $N \geq 2$, $N - 1 \geq N/2$, so

$$\sum_{N \in \mathcal{S}_{2,3}, N \geq 2} \frac{1}{N-1} \leq 2 \sum_{\substack{a,b \geq 0 \\ (a,b) \neq (0,0)}} \frac{1}{2^a 3^b} = 2 \left(\frac{1}{1 - 1/2} \frac{1}{1 - 1/3} - 1 \right) = 4.$$

Unique factorization ensures that the parametrization $N = 2^a 3^b$ has no repetitions. $\square$

8.2 Why a density theorem is the right audit

The classical Müntz–Szász theorem says, in the present integer-exponent setting, that

$$\text{span}\{1, q^m : m \in \Lambda\}$$

is dense in $C[0, 1]$ precisely when $\sum_{m \in \Lambda} 1/m$ diverges. Full versions give the same convergent-sum obstruction in L^p and on intervals away from the origin [2, 4]. Lemma 8.1 therefore implies that the 3-smooth monomials leave an infinite-dimensional blind subspace.

The kernels K_m differ from $1 - q^m$ only by multiplication by $(1 - q)^{-1}$. On any compact subinterval of $(0, 1)$ this multiplier is bounded and invertible. The same incompleteness therefore holds for the complete observation family in (38).

9 Müntz invisibility: an infinite-dimensional no-go theorem

The next theorem is the strongest conclusion of this report. It says that even *all* linear 3-smooth observations, together with the exact endpoint behavior of the measure, do not identify μ_α among positive measures.

Theorem 9.1 (Bounded invisible perturbations). *Fix $0 < \alpha < 1$ and a closed interval $I = [a, b] \subset (0, 1)$. There exists a nonzero bounded real function $g \in L^\infty(\mu_\alpha)$ supported in I such that*

$$\int_0^1 K_m(q)g(q) \, d\mu_\alpha(q) = 0 \quad \text{for every } m \in \Lambda. \quad (41)$$

Consequently, for every sufficiently small $\varepsilon > 0$, the two measures

$$d\mu_\alpha^\pm = (1 \pm \varepsilon g) \, d\mu_\alpha \quad (42)$$

are distinct positive measures satisfying

$$\int_0^1 K_m \, d\mu_\alpha^+ = \int_0^1 K_m \, d\mu_\alpha^- = \int_0^1 K_m \, d\mu_\alpha \quad (m \in \Lambda). \quad (43)$$

They agree with μ_α outside I .

Proof. On I , multiplication by $1 - q$ is a bounded invertible map on $L^1(I)$. If $\text{span}\{K_m : m \in \Lambda\}$ were dense in $L^1(I)$, then $\text{span}\{1 - q^m : m \in \Lambda\}$ would be dense as well, and adding constants would make $\text{span}\{1, q^m : m \in \Lambda\}$ dense. This contradicts the convergent-sum part of the full Müntz theorem and Lemma 8.1. Hence the K_m span is a proper subspace of $L^1(I)$.

By Hahn–Banach and the identification $(L^1(I))^* = L^\infty(I)$, there is a nonzero real $h \in L^\infty(I)$ such that

$$\int_I K_m(q)h(q) \, dq = 0 \quad (m \in \Lambda). \quad (44)$$

The density w_α in (8) is continuous and bounded above and below by positive constants on I . Define

$$g(q) = \begin{cases} h(q)/w_\alpha(q), & q \in I, \\ 0, & q \notin I. \end{cases}$$

Then $g \in L^\infty(\mu_\alpha)$, is nonzero, and (41) follows from (44). Choose $\varepsilon \|g\|_\infty < 1$. Then $1 \pm \varepsilon g > 0$ almost everywhere, so (42) defines distinct positive measures. Equation (43) is immediate from (41), and the support condition gives agreement outside I . $\square$

Corollary 9.2 (Endpoint asymptotics are also invisible). *The perturbations in Theorem 9.1 can be chosen to preserve the density of μ_α on neighborhoods of both 0 and 1. Their ordinary moments differ from those of μ_α by $O(b^j)$, while*

$$\int q^j d\mu_\alpha(q) \sim \alpha j^{\alpha-1}.$$

Thus they have the same polynomial moment asymptotics, to an error exponentially smaller than the main term, and still agree on every 3-smooth observation (38).

Proof. Choose $I = [a, b]$ with $0 < a < b < 1$. The measures coincide off I . Moreover

$$\left| \int_0^1 q^j d(\mu_\alpha^+ - \mu_\alpha^-)(q) \right| \leq 2\varepsilon \|g\|_\infty \mu_\alpha(I) b^j.$$

Combine this with (14). □

Corollary 9.3 (Linear measure reconstruction cannot prove Alaoglu–Erdős). *No argument whose only inputs are*

- (a) *positivity of an unknown measure on $[0, 1]$;*
- (b) *the complete set of scalar observations (38); and*
- (c) *the local endpoint form or polynomial tail asymptotics of μ_α*

can uniquely recover the explicit Hausdorff measure. In particular, linear functional analysis, positive kernels, and Müntz-style measure determination alone cannot settle the Alaoglu–Erdős conjecture.

Remark 9.4. The theorem does not create another exponent β with the same rational powers. The perturbed measures need not belong to the one-parameter family $\{\mu_\beta\}$. Its force is methodological and exact: the missing proof must exploit the *parametric formula* for w_α and the arithmetic of its values nonlinearly, because all linear observation data can be held fixed under positive perturbation.

10 An analytic reformulation of the invisible directions

For a bounded perturbation g supported in $[a, b] \subset (0, 1)$, define its Mellin transform relative to μ_α by

$$G(z) = \int_0^1 q^z g(q) d\mu_\alpha(q). \tag{45}$$

Because the support stays away from 0, G is an entire function of z and decays exponentially along the positive real axis. Condition (41) becomes

$$\sum_{j=0}^{m-1} G(j) = 0 \quad (m \in \Lambda). \tag{46}$$

Thus Müntz incompleteness produces nonzero entire Mellin transforms whose discrete partial sums vanish at every 3-smooth-minus-one index. The zero set is sparse enough to permit this without forcing $G = 0$.

By contrast, the unperturbed transform is explicit:

$$M_\alpha(z) = \int_0^1 q^z d\mu_\alpha(q) = (z+2)^\alpha - (z+1)^\alpha.$$

A future proof would have to combine this special fractional-power form with the arithmetic statement that

$$1 + \sum_{j=0}^{N-2} M_\alpha(j) = N^\alpha = u^a v^b \quad (N = 2^a 3^b)$$

has prime-separated rational denominators. Neither the zero set in (46) nor positivity alone detects that nonlinear coupling.

11 What would constitute a proof along this route?

11.1 The exact nonlinear target

The moment representation converts the conjecture into the following statement.

Research target 11.1 (Arithmetic Hausdorff rigidity). *Prove that no $\alpha \in (0, 1)$ can simultaneously satisfy*

$$2^\alpha \in \mathbb{Z}[1/2], \quad 3^\alpha \in \mathbb{Z}[1/3], \quad (47)$$

using a property of the explicit density w_α that is nonlinear in the observation values

$$\int K_{2^a 3^b - 1} d\mu_\alpha = 2^{a\alpha} 3^{b\alpha} - 1.$$

A sufficient theorem must remain sensitive to the separate 2-primary and 3-primary reduced denominators and cannot depend only on linear combinations of the kernels K_m .

The target is equivalent to the original problem after the fractional normalization, but the report’s no-go theorems sharply restrict plausible proofs. A successful invariant could take one of the following forms:

1. a nonlinear determinant or resultant built from several *reduced* observations, with a lower bound from coprimality that exceeds a sharp analytic upper bound;
2. a rigidity theorem for the explicit Mellin family $(z+2)^\alpha - (z+1)^\alpha$ under coherent dyadic/triadic rational partial sums;
3. a mixed real–nonarchimedean identity that transports the exponent α itself, rather than merely the rational values, to 2-adic and 3-adic analytic data;
4. an unconditional logarithm-product independence theorem strong enough to rule out (1).

The first three are genuinely adapted to the additive moment formulation. The fourth returns to the classical four-exponentials frontier.

11.2 Why the current exact defects do not close

The new defects have three desirable features: positivity, rationality, and exact prime-power denominators. What they lack is an upper bound small enough to force their positive integer numerators to vanish. The natural analytic upper bounds deteriorate only polynomially in the endpoint distance, while the denominator lower bounds are exponentially small in $r + s$. Lemma 3.1 already supplies comparable or stronger one-coordinate bounds. Increasing the finite order cannot change this diagnosis because Corollary 7.3 constructs rational atomic models for every finite collection.

The infinite collection also fails as a linear uniqueness set by Theorem 9.1. Therefore the missing gain cannot be “more positivity” or “more moments” in the same linear sense. It must be a nonlinear arithmetic identity that has no analogue for the rational quadrature and invisible perturbation models.

11.3 A precise conditional resolution

For completeness, the conjecture does have an immediate conditional proof.

Theorem 11.2 (Conditional resolution). *The Alaoglu–Erdős conjecture follows from the Four Exponentials Conjecture, and hence from Schanuel’s conjecture.*

Proof. This is Proposition 2.1(iv). Schanuel’s conjecture implies Four Exponentials by the standard transcendence-degree argument. $\square$

This conditional theorem is not a substitute for an unconditional proof, but it identifies the known transcendence strength sufficient to close every remaining case.

12 Comparison with the supplied report

The contribution of this continuation can be summarized without restating the existing unified report.

Table 2: What is inherited, what is new, and what remains.

Topic	Supplied unified report	This continuation
Four-exponentials barrier	Exhaustive logarithmic, matrix, and determinant formulations.	Only the minimal standalone reduction is recalled.
Conditional counterexample structure	Exact free-monoid and valuation classifications.	Not repeated; fractional normalization is used only to expose u and v .
Additive relation $3 = 2 + 1$	Appears indirectly in finite differences and Loewner-type methods.	Converted into an explicit Hausdorff moment measure for all consecutive increments.

Topic	Supplied unified report	This continuation
Adjacent 3-smooth inputs	No Hausdorff-moment classification.	Exact list $(1, 2), (2, 3), (3, 4), (8, 9)$ and rational moments d_0, d_1, d_2, d_7 .
Arithmetic positivity	Many determinant and valuation certificates.	Exact reduced denominators for one Hankel and three nonconsecutive log-convex defects.
Finite positivity barrier	General determinant-budget obstructions.	Explicit rational two-atom model and general rational finite quadrature theorem.
Infinite linear barrier	Polynomial approximation and spectral sparsity analyses.	Müntz incompleteness gives bounded positive perturbations preserving all 3-smooth observations and endpoint asymptotics.
Final status	Conjecture open; remaining wall isolated.	Conjecture still open; the moment route is narrowed to nonlinear prime-separated arithmetic.

13 Conclusion

The requested unconditional implication has not been proved. The investigation nevertheless produces a coherent new theory around the additive structure that the standard logarithmic matrix forgets.

A hypothetical noninteger solution gives a fractional exponent $0 < \alpha < 1$ for which the increments of t^α form an explicit Hausdorff moment sequence. The only consecutive 3-smooth pairs isolate four rational moments, and those moments generate positive Hankel and log-convexity defects whose reduced denominators can be computed exactly; their numerators are positive and coprime to the displayed denominators (and to 6 when $s > 0$). These are rigorous new arithmetic certificates.

The same analysis also explains why the certificates do not constitute a proof. The first three moments possess an explicit rational two-atom model. Every finite collection of rational polynomial observations has a rational positive quadrature model. Most decisively, the 3-smooth-minus-one exponents have a convergent reciprocal sum, so the full Müntz theorem creates bounded perturbations of the actual measure that preserve all 3-smooth linear data, remain positive, and leave the endpoint singularities and polynomial moment tail unchanged.

The remaining problem is therefore not a generic moment problem. It is a nonlinear arithmetic rigidity problem for the special one-parameter density w_α , with coherent multiplicative values and mutually separated dyadic and triadic denominators. Four Exponentials would settle it immediately. Without that conjecture, a new theorem of precisely this nonlinear type is required.

A Algebraic audit of the exact denominator formulas

For convenient verification, the common-denominator calculations are collected here. Starting from (19),

$$\begin{aligned}
 d_0 d_2 - d_1^2 &= \frac{A_0 A_2}{2^3 r 3^s} - \frac{A_1^2}{2^{2r} 3^{2s}} \\
 &= \frac{3^s A_0 A_2 - 2^r A_1^2}{2^{3r} 3^{2s}}, \\
 d_0^6 d_7 - d_1^7 &= \frac{A_0^6 A_7}{2^{9r} 3^{2s}} - \frac{A_1^7}{2^{7r} 3^{7s}} \\
 &= \frac{3^{5s} A_0^6 A_7 - 2^{2r} A_1^7}{2^{9r} 3^{7s}}, \\
 d_1^5 d_7 - d_2^6 &= \frac{A_1^5 A_7}{2^{8r} 3^{7s}} - \frac{A_2^6}{2^{12r} 3^{6s}} \\
 &= \frac{2^{4r} A_1^5 A_7 - 3^s A_2^6}{2^{12r} 3^{7s}}, \\
 d_0^5 d_7^2 - d_2^7 &= \frac{A_0^5 A_7^2}{2^{11r} 3^{4s}} - \frac{A_2^7}{2^{14r} 3^{7s}} \\
 &= \frac{2^{3r} 3^{3s} A_0^5 A_7^2 - A_2^7}{2^{14r} 3^{7s}}.
 \end{aligned}$$

The parity and residue arguments in Theorems 6.1 and 6.2 show that these common denominators are already reduced.

B A moment-cone interpretation

For a finite set of exponents $E = \{e_1, \dots, e_m\}$, define the moment curve

$$\Phi_E(q) = (q^{e_1}, \dots, q^{e_m}) \in \mathbb{R}^m.$$

The moment vector of μ_α is

$$y_E = \int_0^1 \Phi_E(q) d\mu_\alpha(q).$$

Because the density is positive, y_E lies in the interior of the cone generated by $\Phi_E((0, 1))$. This is the geometric reason rational quadrature works: when y_E happens to be rational, rational points of the moment curve can be perturbed to surround it, and the resulting positive barycentric coordinates can be chosen rational. The transcendental parameter α is therefore invisible to every finite-dimensional moment cone.

For $E = \{0, 1, 2\}$, the cone is described by the truncated Hausdorff conditions

$$m_0 \geq m_1 \geq m_2 \geq 0, \quad m_0 m_2 \geq m_1^2.$$

The sequence from Theorem 4.1 lies strictly in the interior. Formula (34) is an explicit rational point of the fiber over the same moment vector.

C Classical inputs and bibliography notes

The report uses three deep external results.

1. **Gelfond–Schneider.** If a and b are algebraic, $a \neq 0, 1$, and b is algebraic irrational, then every value of a^b is transcendental. It rules out algebraic irrational counterexamples.
2. **Catalan–Mihăilescu.** The only solution in integers $X, Y, m, n > 1$ of $X^m - Y^n = 1$ is $3^2 - 2^3 = 1$. It classifies the consecutive 3-smooth pairs.
3. **Full Müntz theorem.** For a sequence of distinct positive exponents bounded away from 0, convergence of the reciprocal sum prevents density of the associated monomials in C and L^p spaces, including intervals away from the origin. It produces the invisible directions in Theorem 9.1.

No unproved transcendence conjecture is used in the unconditional theorems. Four Exponentials and Schanuel appear only in the explicitly conditional Theorem 11.2.

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